Continuity and the Intermediate Value Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A rectangular field is to be enclosed by a fence. One side of the field is adjacent to a river and does not require fencing. The farmer has 120 meters of fencing material. Let the dimensions of the field be x meters perpendicular to the river and y meters parallel to the river. The area of the field is given by $A = xy$. Part A: Express the area A as a function of x. Part B: Determine the dimensions x and y that maximize the area of the field. Provide the maximum possible area.

Part A: The total length of fencing is $2x + y = 120$. Thus, $y = 120 - 2x$. Substituting this into the area formula, we get $A(x) = x(120 - 2x) = 120x - 2x^2$. Part B: To find the maximum area, we need to find the critical points of $A(x)$ by taking the derivative and setting it to zero. $A'(x) = \frac{dA}{dx} = 120 - 4x$. Setting $A'(x) = 0$: $120 - 4x = 0 \Rightarrow 4x = 120 \Rightarrow x = 30$ meters. To confirm this is a maximum, we can use the second derivative test: $A''(x) = \frac{d^2A}{dx^2} = -4$. Si.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $f(x) = 4$. What is the value of the definite integral $\int_3^6 f(x)\,dx$?

The definite integral $\int_a^b c\,dx$ where c is a constant represents the area of a rectangle with height c and width $b - a$. In this case, $c = 4$, $a = 3$, and $b = 6$. Therefore, the value of the integral is $4 \times (6 - 3) = 4 \times 3 = 12$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

Let the vertex of the inscribed rectangle on the curve be (x, y). Since the vertex is on $y = 9 - x^2$, and the rectangle is in the first quadrant, $x \geq 0$ and $y \geq 0$. The dimensions of the rectangle are x and y. The area of the rectangle is $A = xy$. Substituting $y = 9 - x^2$, we get $A(x) = x(9 - x^2) = 9x - x^3$ for $0 \leq x \leq 3$. To maximize the area, we find the derivative of $A(x)$ with respect to x and set it to zero: $A'(x) = 9 - 3x^2$ Setting $A'(x) = 0$ gives $9 - 3x^2 = 0$, so.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

The net displacement of the first particle from $t = 1$ to $t = 3$ is given by the definite integral of its velocity function over this interval: $\Delta s = \int_1^3 v(t)\,dt$. \\ We are given $v(t) = 3t^2 - 12t + 9$. \\ To find the definite integral, we first find the antiderivative of $v(t)$: \\ $s(t) = \int (3t^2 - 12t + 9)\,dt = t^3 - 6t^2 + 9t + C$. \\ Since the initial position is $s(0) = 5$, we have $0^3 - 6(0)^2 + 9(0) + C = 5$, which means $C = 5$. Thus, $s(t) = t^3 - 6t^2 + 9t + 5$. \\ The net displace.

5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x)$ defined by $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. Let $g(x)$ be a function such that $g(x) = \{f(x), \text{if } x < c; k, \text{if } x \geq c$. If $g(x)$ is continuous at $x = c$ and the limit of $g(x)$ as x approaches c from the left is 5, determine the value of k.

For $g(x)$ to be continuous at $x = c$, the limit from the left must equal the limit from the right, and this must equal the function value at $x = c$. That is, $\lim_{x \to c^-} g(x) = \lim_{x \to c^+} g(x) = g(c)$. We are given that $\lim_{x \to c^-} g(x) = 5$. For $x < c$, $g(x) = f(x) = \frac{x^2 - 9}{x - 3}$. We can simplify this expression for $x \neq 3$: $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. Therefore, $\lim_{x \to c^-} g(x) = \lim_{x \to c^-} (x + 3)$. Since the limit from.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

The differential equation is given by $\frac{dy}{dx} = ky$. This is a separable differential equation. We can rewrite it as $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides, we have $|y| = e^{kx + C_1} = e^{C_1} e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. We are given that the solution curve passes through the point $(0, 1)$. Substituting these values into the general solution: $1 = Ce^{k(0)}$ $1 = Ce^0$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $y = g(x)$ and its inverse $x = h(y)$. The tangent to the graph of $g(x)$ at $x = a$ has a gradient of $\frac{1}{3}$. If $g(a) = 5$, what is the gradient of the tangent to the graph of $h(y)$ at $y = 5$?

We are given that $y = g(x)$ and $x = h(y)$ are inverse functions. The gradient of the tangent to the graph of $g(x)$ at $x = a$ is given by $g'(a) = \frac{dy}{dx}|_{x = a}$. We are told that $g'(a) = \frac{1}{3}$. We are also given that $g(a) = 5$. Since $x = h(y)$ is the inverse function, it means that $h(g(x)) = x$ and $g(h(y)) = y$. The derivative of the inverse function $h(y)$ with respect to y is given by the formula $h'(y) = \frac{1}{g'(x)}$, where $y = g(x)$. We want to find the gradient of the tangent to th.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

The derivative of $f(x)$ is given as $f'(x) = 3x^2 - 12x + 9$. To find the function $f(x)$, we integrate $f'(x)$: $f(x) = \int (3x^2 - 12x + 9)\,dx = x^3 - 6x^2 + 9x + C$. We are given that $f(x)$ passes through $(0, 5)$. Substituting $x = 0$ and $f(0) = 5$ into the expression for $f(x)$: $5 = (0)^3 - 6(0)^2 + 9(0) + C$, which gives $C = 5$. Thus, the function is $f(x) = x^3 - 6x^2 + 9x + 5$. The derivative at $x = 0$ is $f'(0) = 3(0)^2 - 12(0) + 9 = 9$. The point on the graph at $x = 0$ is $(0, f(0)) = (0, 5)$. The equa.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A differentiable function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = 3x - 2$. If $f(1) = 5$, what is the value of $f(3)$?

The given differential equation is $\frac{dy}{dx} = 3x - 2$. To find $f(x)$, we need to integrate $\frac{dy}{dx}$ with respect to x. $y = f(x) = \int (3x - 2)\,dx$. $f(x) = \frac{3}{2}x^2 - 2x + C$, where C is the constant of integration. We are given that $f(1) = 5$. Substituting $x = 1$ into the equation for $f(x)$: $5 = \frac{3}{2}(1)^2 - 2(1) + C$. $5 = \frac{3}{2} - 2 + C$. $5 = \frac{3}{2} - \frac{4}{2} + C$. $5 = -\frac{1}{2} + C$. $C = 5 + \frac{1}{2} = \frac{10}{2} + \frac{1}{2} = \frac{11}{2}$. So, th.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y - 1)^2\cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.

Separate variables: $\frac{dy}{(y-1)^2} = \cos(x)\,dx$. Integrate both sides: $-\frac{1}{y-1} = \sin(x) + C$. Apply initial condition $x = 0, y = 2$: $-\frac{1}{2-1} = \sin(0) + C \Rightarrow -1 = C$. Substitute $C = -1$:

$-\frac{1}{y-1} = \sin(x) - 1 \Rightarrow \frac{1}{y-1} = 1 - \sin(x) \Rightarrow y - 1 = \frac{1}{1-\sin(x)} \Rightarrow \mathbf{y = 1 + \frac{1}{1-\sin(x)}}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?

Let $W(t)$ be the amount of water. $W'(t) = R(t) - P(t)$. A relative maximum occurs when $W'(t) = 0$ and changes from positive to negative, which occurs when $R(t) = P(t)$ and $R(t)$ crosses below $P(t)$.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

The exam consists of 45 Multiple-Choice Questions and 6 Free-Response Questions (51 total). Free Response accounts for exactly 50% of the composite score (54 out of 108 scaled points).

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

The student computed only the net change $\int_0^5 (R(t) - D(t))\,dt$, failing to add the initial condition $W(0) = 100$. By the Fundamental Theorem of Calculus: $W(5) = W(0) + \int_0^5 (R(t) - D(t))\,dt = 100 + 42.18 = \mathbf{142.18}$ gallons. Failing to include the initial condition is a common fatal error on FRQ Question 1.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Definition of the Derivative

Level 2 · MediumWhy does College Board recommend storing intermediate values in graphing calculator memory variables rather than writing down a 3-decimal rounded number and reusing it?

Reusing intermediate numbers rounded to 3 decimal places often propagates rounding errors that cause the final computed answer to be off by 0.001 or more, forfeiting the final answer point on Free Response.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Level 3 · AdvancedA student writes: 'It is increasing because it's derivative is positive.' Why will this response lose credit on an AP exam?

College Board readers deduct points for ambiguous pronouns ('it'). The student must explicitly identify the subject: 'The function $f(x)$ is increasing because $f'(x) > 0$.'

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.01 The Chain Rule**

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

Write out the terms: $f(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \ldots$ Recall the definition of the hyperbolic sine function: $\sinh(x) = \frac{e^x - e^{-x}}{2}$. Expanding e^x and e^{-x} shows that only odd powers remain with coefficients $1/(2n+1)!$. Thus $f(x) = \sinh\mathbf{(x)}$. Apply the Ratio Test: \[\lim_{n\to\infty} \frac{x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{x^{2n+1}} = x^2 \lim_{n\to\infty} \frac{1}{(2n+3)(2n+2)} = 0

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

Acceleration is the derivative of velocity: $\vec{a}(3) = \langle x''(3), y''(3) \rangle$. Use the calculator's numerical derivative function nDeriv: • $x''(3) = \frac{d}{dt}[\ln(t^2+1)]\Big|_{t=3} = \frac{2(3)}{3^2+1} = \frac{6}{10} = 0.6$. • $y''(3) = \frac{d}{dt}[3\cos(t)]\Big|_{t=3} = -3\sin(3) \approx -0.423$. Thus: $\mathbf{\vec{a}(3) = \langle 0.600, -0.423 \rangle}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 1 · FoundationsFind all values of c in the open interval $(-1, 2)$ that satisfy the Mean Value Theorem for $f(x) = x^3 - 2x^2 + 1$.

1. Check hypotheses: $f(x)$ is a polynomial, continuous on $[-1, 2]$ and differentiable on $(-1, 2)$. 2. Calculate the average rate of change on $[-1, 2]$: $f(-1) = (-1)^3 - 2(-1)^2 + 1 = -1 - 2 + 1 = -2$ $f(2) = 2^3 - 2(2)^2 + 1 = 8 - 8 + 1 = 1$ $\frac{f(2) - f(-1)}{2 - (-1)} = \frac{1 - (-2)}{3} = \frac{3}{3} = 1$ 3. Set $f'(c) = 1$: $f'(x) = 3x^2 - 4x \Rightarrow 3c^2 - 4c = 1 \Rightarrow 3c^2 - 4c - 1 = 0$ Apply the quadratic formula.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ **is defined,** $\lim_{x \to c} f(x)$ **exists, and** $\lim_{x \to c} f(x) = f(c)$ ✓

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $f(c)$ is continuous on (a, b), $f(c) = 0$, and $f'(c) \neq 0$

d. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

By the formal definition of continuity at a point, $f(c)$ must be defined, the two-sided limit must exist, and the limit must equal the function value.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make f continuous at $x = 3$?

a. 3

b. 6 ✓

c. 0

d. No value can make it continuous

The limit is $\lim_{x \to 3} \frac{(x - 3)(x + 3)}{x - 3} = 3 + 3 = 6$. Therefore, $f(3)$ must equal 6.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?

a. **Infinite (essential) discontinuity** ✓

b. Removable discontinuity

c. Jump discontinuity

d. Oscillating discontinuity

Because $\lim_{x \to 2^+} g(x) = \infty$ and $\lim_{x \to 2^-} g(x) = -\infty$, there is an infinite discontinuity (vertical asymptote) at $x = 2$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

a. 1 ✓

b. 2

c. 3

d. 0

Equate left and right limits at $x = 2$: $k(2^2) + 1 = 3(2) - 1 \Rightarrow 4k + 1 = 5 \Rightarrow 4k = 4 \Rightarrow k = 1$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of a ensures continuity at $x = 0$?

a. 1

b. 2

c. 4 ✓

d. $\frac{1}{4}$

$\lim_{x \to 0} \frac{\sin(ax)}{x} = a$. For continuity, $a = g(0) = 4$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

a. **Continuous because both one-sided limits equal** 3 ✓

b. Jump discontinuity of size 1

c. Infinite discontinuity

d. Undefined at $x = 1$

$\lim_{x \to 1^-} (x + 2) = 3$ and $\lim_{x \to 1^+} (2x + 1) = 3 = h(1)$. Therefore, h is continuous at $x = 1$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the hypotheses of the Intermediate Value Theorem for a function f on interval $[a, b]$?

a. f **must be continuous on the closed interval** $[a, b]$ ✓

b. f must be differentiable on (a, b)

c. $f(a)$ and $f(b)$ must have opposite signs

d. f must be strictly increasing on $[a, b]$

The Intermediate Value Theorem requires only that f is continuous on the closed interval $[a, b]$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let f be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?

a. Yes, because 0 lies between -3 and 7 ✓

b. No, f must also be differentiable

c. No, f must be a polynomial

d. Only if $f'(x) > 0$ for all x

Because f is continuous on $[1, 5]$ and $f(1) < 0 < f(5)$, the IVT guarantees at least one $c \in (1, 5)$ such that $f(c) = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

a. $[1, 2]$ ✓

b. $[0, 1]$

c. $[-1, 0]$

d. $[2, 3]$

$P(1) = 1 - 3 - 1 = -3 < 0$, and $P(2) = 8 - 6 - 1 = 1 > 0$. By IVT, a zero exists in $(1, 2)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What does the Extreme Value Theorem guarantee for a function f continuous on a closed interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$ ✓

b. f has a horizontal tangent on (a, b)

c. $f'(c) = 0$ for some $c \in (a, b)$

d. f is invertible on $[a, b]$

The EVT states that any continuous function on a closed bounded interval $[a, b]$ must attain an absolute maximum and an absolute minimum.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

a. The interval is open at 0 and f increases without bound as $x \to 0^+$ ✓

b. f is not differentiable on $(0, 2]$

c. f is bounded on $(0, 2]$

d. $f(2) = 0.5$ is the maximum

The interval $(0, 2]$ is not closed, so the EVT does not apply. As $x \to 0^+$, $f(x) \to \infty$, so no absolute maximum exists.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

a. 5 ✓

b. 4

c. 1

d. 6

Critical point: $f'(x) = -2x + 4 = 0 \Rightarrow x = 2$. Evaluate: $f(0) = 1, f(2) = -(4) + 8 + 1 = 5, f(3) = -(9) + 12 + 1 = 4$. Absolute maximum is 5.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of a function $g(x)$ is shown above. The function $g(x)$ is defined by $g(x) = \frac{1}{3}x^3 - ax^2 + 2x + 1$, where a is a positive constant. The graph has a tangent line at $x = 1$ which is parallel to the line $y = -2x + 5$. Find the value of the constant a.

a. 1

b. 2

c. 3 ✓

d. 4

The derivative of $g(x)$ is $g'(x) = x^2 - 2ax + 2$. The slope of the tangent line at $x = 1$ is equal to the derivative of $g(x)$ evaluated at $x = 1$. The given tangent line is $y = -2x + 5$, so its slope is -2. Therefore, $g'(1) = -2$. Substituting $x = 1$ into the derivative: $g'(1) = (1)^2 - 2a(1) + 2 = 1 - 2a + 2 = 3 - 2a$. Setting $g'(1)$ equal to the slope of the tangent line: $3 - 2a = -2$. Solving for a: $-2a = -5$, which gives $a = \frac{5}{2}$. However, upon review, the problem asks for a positive constan.

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x)$ be a function such that $f'(x) = \cos(x^2)$ and $g(x)$ be a function defined by $g(x) = \sqrt{x}$. ; ; \textbf{(a)} Find the derivative of the composite function $h(x) = f(g(x))$. ; ; \textbf{(b)} If $f(1) = \pi$, find the value of $h'(4)$.

a. $\frac{\cos(4)}{2}$ ✓

b. $\frac{\pi\cos(4)}{2}$

c. $\frac{\cos(16)}{2}$

d. $\frac{\pi\cos(16)}{2}$

\textbf{(a)} We use the chain rule to find the derivative of $h(x) = f(g(x))$. ; The chain rule states that $h'(x) = f'(g(x)) \cdot g'(x)$. ; We are given $f'(x) = \cos(x^2)$ and $g(x) = \sqrt{x}$. ; First, we find $g'(x)$: $g'(x) = \frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$. ; Next, we find $f'(g(x))$ by substituting $g(x)$ into $f'(x)$: $f'(g(x)) = \cos((\sqrt{x})^2) = \cos(x)$. ; Therefore.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the value of $\lim_{h\to 0} \frac{(2+h)^4 - 16}{h}$?

a. 8

b. 16

c. **32** ✓

d. 64

By definition, the derivative of a function f at $x = a$ is given by: $f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$ The expression $\lim_{h\to 0} \frac{(2+h)^4 - 16}{h}$ matches this definition for the function $f(x) = x^4$ evaluated at $a = 2$, since $f(2) = 2^4 = 16$. Applying the power rule to find the derivative: $f'(x) = 4x^3$ Evaluating at $x = 2$: $f'(2) = 4(2)^3 = 4(8) = 32$ Common Misconceptions: - Choice A (8): Results from incorrectly applying the power rule as $4x$ and evaluating a.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the function $f(x) = x^2 - 3x$. If $g(x)$ is a function such that $h(x) = f(g(x))$, and at $x = 2$, $g(2) = 5$ and $g'(2) = 3$, find $h'(2)$.

a. **21** ✓

b. 30

c. 15

d. 6

This question involves the chain rule for differentiation. The chain rule states that if $h(x) = f(g(x))$, then $h'(x) = f'(g(x)) \cdot g'(x)$. First, we find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 3x) = 2x - 3$. Now, we need to find $f'(g(x))$. Since $g(x)$ is a general function, we substitute $g(x)$ into $f'(x)$: $f'(g(x)) = 2g(x) - 3$. To find $h'(2)$, we apply the chain rule formula at $x = 2$: $h'(2) = f'(g(2)) \cdot g'(2)$. We are given that $g(2) = 5$ and $g'(2) = 3$. Substitute $g(2) = 5$ into.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x) = \sin(x^2)$. What is the derivative of $f(x)$?

a. $\cos(x^2)$

b. $2x\cos(x^2)$ ✓

c. $\sin(2x)$

d. $-2x\cos(x^2)$

To find the derivative of $f(x) = \sin(x^2)$, we use the chain rule. The chain rule states that if $f(x) = g(h(x))$, then $f'(x) = g'(h(x)) \cdot h'(x)$. In this case, let $g(u) = \sin(u)$ and $h(x) = x^2$. Then $g'(u) = \cos(u)$ and $h'(x) = 2x$. Applying the chain rule, we get $f'(x) = \cos(x^2) \cdot 2x$, which simplifies to $2x\cos(x^2)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ **is always positive and increasing.** ✓

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

The velocity of the particle is given by $v(t) = \sqrt{t^2 + 1}$. Since $t \geq 0$, $t^2 \geq 0$, so $t^2 + 1 \geq 1$. Therefore, $v(t) = \sqrt{t^2 + 1} \geq \sqrt{1} = 1$ for all $t \geq 0$. This means the velocity is always positive. The position of the particle is given by $x(t)$. The Fundamental Theorem of Calculus states that the change in position (displacement) is the integral of the velocity function: $x(t) - x(0) = \int_0^t v(u)\,du$. The function $P(x)$ represents the total displacement from $t = 0$ to.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

a. $t = \frac{3}{2}$ ✓

b. $t = 1$

c. $t = 2$

d. $t = 3$

To find the acceleration $a(t)$, take the derivative of the velocity function $v(t)$ with respect to time t: $a(t) = v'(t) = \frac{d}{dt}(6t^2 - 18t + 12) = 12t - 18$. Set $a(t) = 0$ and solve for t: $12t - 18 = 0 \Rightarrow 12t = 18 \Rightarrow t = \frac{18}{12} = \frac{3}{2}$. Common Misconceptions: - Solving $v(t) = 0$ yields $6(t-1)(t-2) = 0 \Rightarrow t = 1$ or $t = 2$. These are the times when the particle is at rest, not when its acceleration is zero. - Omitting the factor of 2 when applying the po.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. **g'(2) = 2** ✓

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. We need to find $g'(2)$, which means we need to evaluate $f(2)$. From the graph, at $x = 2$, the function $f(x)$ is on a line segment connecting $(0, 4)$ and $(4, 0)$. The equation of this line segment can be found. The slope is $m = \frac{0-4}{4-0} = -1$. Using the point-slope form with $(4, 0)$, we get $y - 0 = -1(x - 4)$, which simplifies to $y = -x + 4$. Therefore, $f(x) = -x + 4$ for $0 \leq x \leq 4$. Evaluating $f(2)$ gives $f(2) = -2 + 4 = 2$..

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

a. 10

b. 15

c. **20** ✓

d. 25

The displacement of the particle is the definite integral of its velocity function over the given time interval. \\ Displacement = $\int_0^4 v(t)\,dt = \int_0^4 5\,dt$. \\ To evaluate this definite integral, we find the antiderivative of 5, which is $5t$. \\ Then, we evaluate the antiderivative at the limits of integration: \\ $[5t]_0^4 = 5(4) - 5(0) = 20 - 0 = 20$. \\ Therefore, the total displacement of the particle is 20 units.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

If $f(x) = x^2 + 3$ is increasing on $[1, 6]$, a left Riemann sum with equal subintervals will:

a. equal the true area exactly

b. **underestimate the true area** ✓

c. overestimate the true area

d. cannot be determined

For an increasing function, left-endpoint heights are too small and right-endpoint heights are too large, so a left sum will underestimate the true area.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

If $f(x) = 2x^2 + 5$ is increasing on $[2, 5]$, a right Riemann sum with equal subintervals will:

a. equal the true area exactly

b. cannot be determined

c. underestimate the true area

d. **overestimate the true area** ✓

For an increasing function, left-endpoint heights are too small and right-endpoint heights are too large, so a right sum will overestimate the true area.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Using a right Riemann sum with 4 subintervals of equal width on $[0, 2]$, approximate $\int_0^2 (x^2 + x - 2)\,dx$.

a. $\frac{21}{4}$

b. $-\frac{3}{4}$

c. $-\frac{15}{4}$

d. $\frac{9}{4}$ ✓

$\Delta x = \frac{1}{2}$; heights at the right endpoints give $\frac{1}{2}(-\frac{5}{4} + 0 + \frac{7}{4} + 4) = \frac{9}{4}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Using the Trapezoidal Rule with 2 subintervals on $[0, 6]$, approximate $\int_0^6 (2x^2 - 3x - 1)\,dx$.

a. 183

b. 21

c. 104

d. **102** ✓

$T = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + f(x_2)] = \frac{3}{2}[-1 + 2(8) + 53] = 102$

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